# CVE-2025-45528 - Nitro PDF Pro Arbitrary Code Execution

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| | | | |
|---|---|---|---|
| **1** Total Advisories | **1** Critical | **0** High | **0** Medium |
| **CRITICAL** Severity | **9.8** CVSS / Risk Score | **TLP:CLEAR** TLP Classification | **PRO** Customer Tier |

## Executive Summary

[P1] Remote Code Execution - moderate severity (Risk 5.9/10). CVSS 9.8/10 severity score. No MITRE ATT&CK techniques mapped - limited adversarial context. 1 indicator detected - upgrade to Pro for full IOC access. Actor attribution: Vulnerability Exploitation Cluster. AI confidence: MODERATE (40%) - partial intelligence corroboration.

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## Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

| Advisory Title | Severity | CVSS | EPSS | AI Summary |
|---|---|---|---|---|
| CVE-2025-45528 - Nitro PDF Pro Arbitrary Code Execution | CRITICAL | 9.8 | - | [P1] Remote Code Execution - moderate severity (Risk 5.9/10). CV |

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## Vulnerability Intelligence Deep-Dive

| 9.8 | N/A | NO | CRITICAL |
|---|---|---|---|
| CVSS 3.1 Score | EPSS 30-day Probability | CISA KEV Listed | Severity Rating |

| Field | Value |
|---|---|
| CVE / Advisory ID | CVE-2025-45528 |
| Vulnerability Class | Unclassified Security Vulnerability |
| CWE Reference | Pending NVD enrichment |
| Actor Attribution | UNC-CDB |
| Source Advisory | NVD / Vendor Advisory |

Refer to the official NVD entry and vendor advisory for the complete list of affected products and versions for CVE-2025-45528.

**Vulnerability Context & Attack Surface**

This vulnerability affects the security boundary of the target component. Successful exploitation may allow unauthorized access, data disclosure, or denial of service depending on deployment configuration. Consult the NVD entry and vendor advisory for full technical root cause analysis and CVSS vector breakdown.

## MITRE ATT&CK® v15 Mapping

Technique density score: **0.0** · Techniques observed: **0** · Tactics covered: **0**

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

## Threat Actor Intelligence

Attribution analysis across advisories in this report period.

| Threat Actor | Count | Associated CVEs / Indicators |
|---|---|---|
| UNC-CDB | 1 | - |

## Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

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## Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

### Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CVE-2025-45528 Exploitation Attempt
id: cdb-cve-2025-45528-det
status: experimental
description: Detects exploitation indicators related to CVE-2025-45528.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/10/24
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CVE-2025-45528'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

### Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - CVE-2025-45528 - Nitro PDF Pro Arbitrary Code Execution
// CVE-2025-45528 | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "CVE-2025-45528"
or AlertName has "CVE-2025-45528"
or ExtendedProperties has "CVE-2025-45528"
| extend Rule = "APEX-CVE202545528", Severity = "CRITICAL"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

## Incident Response Playbook

**Phase 1 - 0-24 Hours (Containment)**

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield CVE instances exposed to the internet.
3. Enable verbose access logging on all CVE endpoints immediately.
4. Pull last 72h of web server and application logs for forensic baselining.

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5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
6. Notify relevant internal stakeholders and legal / compliance team.

### Phase 2 - 24-72 Hours (Eradication)

1. Apply vendor patch for CVE-2025-45528 or implement recommended mitigation.
2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
3. Rotate all API keys, service account credentials, and session tokens.
4. Conduct forensic timeline reconstruction for potential compromise window.
5. Validate patch deployment across 100% of affected instances via asset inventory.

### Phase 3 - 7 Days (Recovery & Hardening)

1. Conduct post-incident review and update incident response runbooks.
2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
3. Update asset inventory with patched version information and closure evidence.
4. Submit findings to internal risk register and CISO dashboard.
5. Schedule follow-up vulnerability scan to confirm remediation completeness.

## Financial Impact Analysis (FAIR Model)

| $2.4M - $12M | $5.8M | $1.2M/day | $850K |
|---|---|---|---|
| Breach Cost Range (FAIR) | IBM Cost of Breach 2025 Median | Business Interruption Cost | Regulatory Fine Exposure |

Critical severity breaches trigger mandatory 72-hour regulatory disclosure (GDPR Art. 33), class-action exposure, and long-tail legal costs averaging 18-36 months post-incident.

**Remediation Cost Estimate:** $380K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

## Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

| Framework | Reference | Obligation |
|---|---|---|
| **GDPR** | Art. 33 / Art. 83(4) | 72-hour breach notification to DPA; fines up to 4% of global annual turnover |
| **PCI-DSS v4.0** | Req. 6.3 / 6.4 | Critical patches within 1 month; compensating controls required |
| **NIS2 Directive** | Art. 23 | Early warning within 24h; full incident report within 72h |
| **SOX / ICFR** | Section 302 / 404 | Material cybersecurity weakness disclosure in 10-K/10-Q |

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| Framework | Reference | Obligation |
|---|---|---|
| **ISO 27001:2022** | A.8.8 / A.5.29 | Documented vulnerability management and incident management |

## Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for CVE-2025-45528 - Nitro PDF Pro Arbitrary Code Execution.
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK techniques.
4. Apply vendor patch for CVE-2025-45528 - Nitro PDF Pro Arbitrary Code Execution immediately - check NVD for official fix guidance.
5. Enable enhanced logging and alerting on all systems running affected software versions.
6. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
7. Audit all internet-facing instances of affected products and confirm patch deployment.

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# Appendix - Report Metadata & Legal

| Field | Value |
|---|---|
| Report ID | intel--98dff366cd298edcf32a4400 |
| Report Type | Weekly |
| Platform | CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE |
| Generated At | 2026-10-24T04:40:34.08729 |
| Customer Tier | PRO |
| Customer Email | - |
| TLP | TLP:CLEAR |
| STIX Version | 2.1 |
| ATT&CK Version | v15 |

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